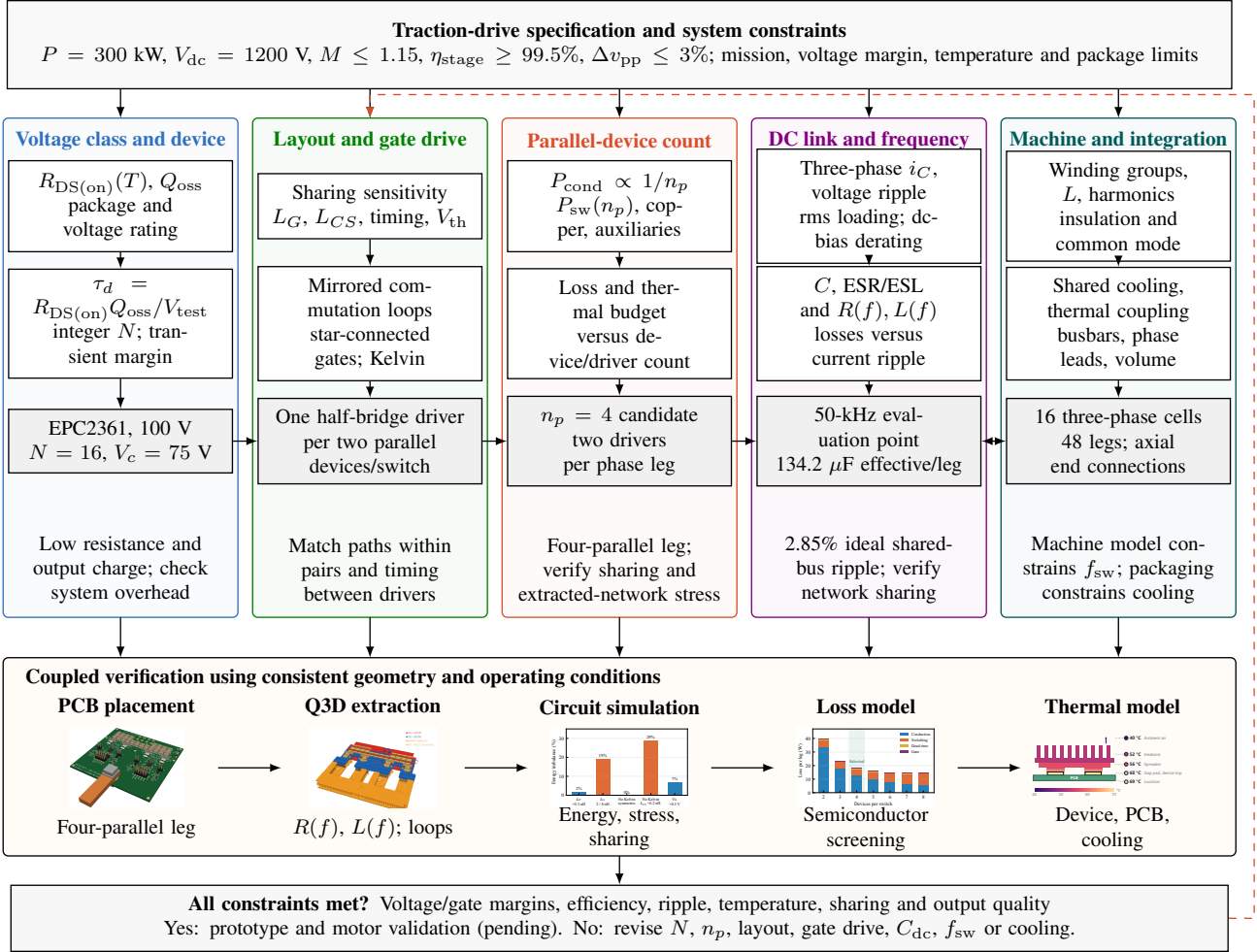

Traction-drive specification and system constraints
$P = 300$ kW, $V_{dc} = 1200$ V, $M \leq 1.15$, $\eta_{stage} \geq 99.5\%$, $\Delta v_{pp} \leq 3\%$; mission, voltage margin, temperature and package limits
Voltage class and device
$R_{DS(on)}(T)$, Q_{oss} package and voltage rating
$\tau_d = R_{DS(on)} Q_{oss}/V_{test}$ integer N; transient margin
EPC2361, 100 V $N = 16$, $V_c = 75$ V
Low resistance and output charge; check system overhead
Layout and gate drive
Sharing sensitivity L_G, L_{CS}, timing, V_{th}
Mirrored commutation loops star-connected gates; Kelvin
One half-bridge driver per two parallel devices/switch
Match paths within pairs and timing between drivers
Parallel-device count
$P_{cond} \propto 1/n_p$ $P_{sw}(n_p)$, copper, auxiliaries
Loss and thermal budget versus device/driver count
$n_p = 4$ candidate two drivers per phase leg
Four-parallel leg; verify sharing and extracted-network stress
DC link and frequency
Three-phase i_C, voltage ripple rms loading; dc-bias derating
C, ESR/ESL and $R(f)$, $L(f)$ losses versus current ripple
50-kHz evaluation point 134.2 μF effective/leg
2.85% ideal shared-bus ripple; verify network sharing
Machine and integration
Winding groups, L, harmonics insulation and common mode
Shared cooling, thermal coupling busbars, phase leads, volume
16 three-phase cells 48 legs; axial end connections
Machine model constrains f_{sw}; packaging constrains cooling
Coupled verification using consistent geometry and operating conditions
PCB placement
Four-parallel leg
Q3D extraction
$R(f)$, $L(f)$; loops
Circuit simulation
Energy, stress, sharing
Loss model
Semiconductor screening
Thermal model
Device, PCB, cooling
All constraints met? Voltage/gate margins, efficiency, ripple, temperature, sharing and output quality
Yes: prototype and motor validation (pending). No: revise N, n_p, layout, gate drive, C_{dc}, f_{sw} or cooling.